

Updated Models and Likelihood Construction

TITE-CRM for Random CRM, Fixed- r CRM, and Alpha-CRM; Delayed-Outcome Likelihood for IPCRM

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1 Purpose

This note documents the updated likelihoods for the current AIDE-CRM helper and JAGS model files. The implementation follows two different time-to-event conventions:

- **Random CRM, fixed- r CRM, and alpha-CRM:** use the **TITE-CRM weighted likelihood**. A patient with partial follow-up contributes through the event probability $w_i p_i$, where w_i is the fraction of the DLT assessment window completed.
- **IPCRM:** use the **delayed-outcome approximation**. A pending non-DLT contributes $(1 - p_i)^{w_i}$, where w_i is the fraction of evaluated time.

The key difference is the pending non-DLT contribution:

TITE-CRM pending non-DLT: $1 - w_i p_i$, Delayed-outcome pending non-DLT: $(1 - p_i)^{w_i}$.

2 Interim-data notation

At an interim decision time, for patient $i = 1, \dots, N$:

Symbol	Meaning
d_i	Assigned dose level, $d_i \in \{1, \dots, J\}$.
z_i	Retreat/IPDE indicator. $z_i = 1$ for IPDE/retreat observations and $z_i = 0$ for regular observations.
y_i	Observed DLT indicator by the decision time. Pending patients are coded $y_i = 0$.
u_i	Follow-up time accumulated by the decision time.
T	Full DLT assessment window.
w_i	Fraction of evaluated time, clipped to $[0, 1]$: $w_i = \min\{1, \max(0, u_i/T)\}$. The code sets $w_i = 1$ for observed DLTs and fully evaluated patients.
q_j	CRM skeleton probability at dose j .

The data preparation code uses an explicit weight column first. Recognized names include `tite_weight`, `weight`, `w`, `wi`, `followup_fraction`, `eval_frac`, `frac_eval`, `fraction_evaluated`, and `followup_prop`. If no explicit column is supplied, the helper can compute w_i from `followup_time / assessment_window` or from `(decision_time - arrival_time) / assessment_window`. Observed DLTs and fully evaluated patients are forced to $w_i = 1$.

3 Source likelihood ideas

The TITE-CRM approach uses a weighted dose-response model

$$G(d, w, \beta) = wF(d, \beta),$$

so the observed interim likelihood contribution is

$$L_i(\beta) = \{w_i F(d_i, \beta)\}^{y_i} \{1 - w_i F(d_i, \beta)\}^{1-y_i}.$$

With $w_i = 1$ for completed observations, this reduces to the ordinary Bernoulli likelihood for fully evaluated patients.

The delayed-outcome approach starts from the observed likelihood for pending DLT data. For a pending patient who has not yet experienced DLT, the exact term is $1 - w_i p_j$, and the model-assisted approximation replaces it with

$$1 - w_i p_j \approx (1 - p_j)^{w_i}.$$

This gives an effective non-DLT contribution that is fractional on the log-likelihood scale.

4 Fixed- r and random CRM: discount model with TITE-CRM likelihood

For the fixed- r and random- r discount CRM backends, the dose-toxicity curve is the power CRM model

$$p_j(\alpha) = q_j^{\exp(\alpha)}, \quad \alpha \sim N(0, \sigma_\alpha^2).$$

For an IPDE/retreat observation at dose j , the carryover-discounted toxicity probability is

$$\theta_j^{\text{IPDE}} = r + (1 - r)p_j(\alpha).$$

For a regular observation, the probability is $p_j(\alpha)$. For a mixed dataset containing both regular and IPDE observations, define

$$\theta_i = (1 - z_i)p_{d_i}(\alpha) + z_i\theta_{d_i}^{\text{IPDE}}.$$

The TITE-CRM likelihood contribution is

$$L_i(\alpha, r) = (w_i \theta_i)^{y_i} (1 - w_i \theta_i)^{1-y_i}.$$

Therefore the full likelihood is

$$L(\alpha, r) = \prod_{i=1}^N (w_i \theta_i)^{y_i} (1 - w_i \theta_i)^{1-y_i}.$$

For the **fixed- r CRM**, r is user-specified, e.g. $r = 0.10$. The posterior is proportional to

$$\pi(\alpha \mid \mathcal{D}) \propto \left\{ \prod_{i=1}^N (w_i \theta_i)^{y_i} (1 - w_i \theta_i)^{1-y_i} \right\} \phi(\alpha; 0, \sigma_\alpha^2).$$

For the **random CRM**, r has a beta prior,

$$r \sim \text{Beta}(a_r, b_r),$$

and the joint posterior is proportional to

$$\pi(\alpha, r \mid \mathcal{D}) \propto \left\{ \prod_{i=1}^N (w_i \theta_i)^{y_i} (1 - w_i \theta_i)^{1-y_i} \right\} \phi(\alpha; 0, \sigma_\alpha^2) \text{Beta}(r; a_r, b_r).$$

JAGS core used in the updated files

```
for (j in 1:J) {
  p[j] <- pow(q[j], exp(alpha))
  theta_ipde[j] <- r + (1 - r) * p[j]
}

for (i in 1:N) {
  theta_base[i] <- (1 - ipde[i]) * p[dose[i]] +
    ipde[i] * theta_ipde[dose[i]]
  theta[i] <- w[i] * theta_base[i]
  y[i] ~ dbern(theta[i])
}
```

The fixed- r model uses `fix_CRM.TITE.bug`; the random- r model uses `random_CRM.TITE.bug`. The only substantive difference is whether r is supplied as data or sampled from `dbeta(a_r, b_r)`.

5 Alpha-CRM: deterministic carryover model with TITE-CRM likelihood

The alpha-CRM backend uses deterministic carryover through an effective cumulative dose. To avoid notational conflict with the CRM parameter, denote the carryover grid value by $a \in (0, 1)$ and the CRM power parameter by θ .

For a patient-cycle row i , the cumulative effective dose is

$$D_i(a) = \sum_{k \leq i, \text{ same patient}} d_k a^{(t_i - t_k)/T},$$

where d_k is the actual/scaled dose value from the previous or current treatment row and $t_i - t_k$ is the elapsed calendar time.

The function $S\{D_i(a)\}$ is obtained by linearly interpolating the skeleton across the dose-value grid. The model probability is

$$p_i(\theta, a) = S\{D_i(a)\}^{\exp(\theta)}, \quad \theta \sim N(\mu_\theta, \sigma_\theta^2).$$

The updated alpha-CRM likelihood uses the same TITE-CRM form:

$$L_i(\theta, a) = \{w_i p_i(\theta, a)\}^{y_i} \{1 - w_i p_i(\theta, a)\}^{1-y_i}.$$

For a fixed grid value a , the integrated marginal likelihood is

$$m(a) = \int \left[\prod_{i=1}^N \{w_i p_i(\theta, a)\}^{y_i} \{1 - w_i p_i(\theta, a)\}^{1-y_i} \right] \phi(\theta; \mu_\theta, \sigma_\theta^2) d\theta.$$

The posterior weight for each carryover grid value is

$$\omega(a) = \frac{m(a)}{\sum_{a' \in \mathcal{A}} m(a')}.$$

Posterior toxicity estimates are then averaged over both θ and the carryover grid:

$$\hat{p}_k = \sum_{a \in \mathcal{A}} \omega(a) \frac{\int q_k^{\exp(\theta)} L(\theta, a) \phi(\theta) d\theta}{\int L(\theta, a) \phi(\theta) d\theta}.$$

R likelihood core used in alpha-CRM

```
p <- S_at_D^exp(theta)
p_eff <- tite_weight * p
sum(y * log(p_eff) + (1 - y) * log1p(-p_eff))
```

This corresponds to

$$\log L(\theta, a) = \sum_i y_i \log(w_i p_i) + (1 - y_i) \log(1 - w_i p_i).$$

6 IPCRM: logistic cumulative model with delayed-outcome likelihood

In the updated helper, `r_model = "ipcrm"` is mapped to the logistic cumulative CRM backend. For row i ,

$$\text{logit}(p_i) = \beta_0 + \beta_1 x_i + \beta_2 c_i,$$

where x_i is the assigned dose score and c_i is the cumulative previous-dose score for that patient. The baseline toxicity probability at dose j is

$$\text{logit}(p_j^{\text{base}}) = \beta_0 + \beta_1 s_j,$$

where s_j is the dose score for dose j .

The delayed-outcome likelihood is

$$L_i(\beta_0, \beta_1, \beta_2) = p_i^{y_i} (1 - p_i)^{(1-y_i)w_i}.$$

This single expression covers the three cases:

Observation status	Code	Likelihood contribution
Observed DLT	$y_i = 1, w_i = 1$	p_i
Fully evaluated non-DLT	$y_i = 0, w_i = 1$	$1 - p_i$
Pending non-DLT	$y_i = 0, 0 < w_i < 1$	$(1 - p_i)^{w_i}$

The corresponding log-likelihood is

$$\ell_i = y_i \log(p_i) + (1 - y_i)w_i \log(1 - p_i).$$

The JAGS implementation uses the zeros trick:

```
logit(p[i]) <- beta0 + beta1 * assigned_d[i] + beta2 * cumu_d[i]
loglik[i] <- y[i] * log(p[i]) + (1 - y[i]) * w[i] * log(1 - p[i])
zeros[i] ~ dpois(-loglik[i])
```

This is intentionally different from TITE-CRM. For a pending non-DLT, IPCRM uses $(1 - p_i)^{w_i}$ rather than $1 - w_i p_i$.

7 Comparison of the updated likelihoods

Backend	Model probability	Pending non-DLT term	DLT term
Fixed- r CRM	$\theta_i = (1 - z_i)p_{d_i} + z_i\{r + (1 - r)p_{d_i}\}$	$1 - w_i\theta_i$	θ_i because $w_i = 1$
Random CRM	Same as fixed- r , but $r \sim \text{Beta}(a_r, b_r)$	$1 - w_i\theta_i$	θ_i because $w_i = 1$
Alpha-CRM	$p_i(\theta, a) = S\{D_i(a)\}^{\exp(\theta)}$	$1 - w_i p_i$	p_i because $w_i = 1$
IPCRM	$\text{logit}(p_i) = \beta_0 + \beta_1 x_i + \beta_2 c_i$	$(1 - p_i)^{w_i}$	p_i

8 Final-selection and overdose-elimination note

For empirical overdose elimination in `select.mtd.crm()`, the updated helper uses an effective sample size based on weighted non-DLT information:

$$y_j = \sum_i I(d_i = j)y_i, \quad n_j^{\text{eff}} = y_j + \sum_{i: d_i=j, y_i=0} w_i.$$

Then the beta-binomial overdose probability is computed as

$$\Pr(p_j > \phi \mid y_j, n_j^{\text{eff}}) = 1 - F_{\text{Beta}(y_j+1, n_j^{\text{eff}}-y_j+1)}(\phi).$$

This keeps the elimination rule from treating pending patients as fully observed non-DLTs.

9 Implementation checklist

1. Pending patients must have $y_i = 0$ and $0 \leq w_i < 1$.
2. Observed DLT patients must have $y_i = 1$ and are forced to $w_i = 1$.
3. Fully evaluated non-DLT patients must have $y_i = 0$ and $w_i = 1$.
4. Use `type = "retreat"` for IPDE/retreat rows in the discount CRM models.
5. Use `r_model = "fixed"` or `"r_fixed"` for fixed- r CRM, `r_model = "random"` for random CRM, `r_model = "alpha_crm"` for alpha-CRM, and `r_model = "ipcrm"` for IPCRM.
6. For fixed/random CRM, the default model files are `fix_CRM.TITE.bug` and `random_CRM.TITE.bug`.
7. For IPCRM, the helper writes the delayed-outcome JAGS model internally when needed.

10 References

- Cheung, Y. K., and Chappell, R. (2000). Sequential designs for phase I clinical trials with late-onset toxicities. *Biometrics*, 56, 1177–1182.
- Lin, R., and Yuan, Y. (2020). Time-to-event model-assisted designs for dose-finding trials with delayed toxicity. *Biostatistics*, 21(4), 807–824.